

Product Requirements Document (PRD)

Project: Event Listing and Registration Platform

Objective

Build a responsive event listing platform where users can browse events, filter them based on category or date, register for events, and view attendee counts. The platform should also provide a simple admin interface to manage events using CRUD operations.

The homepage should additionally include a chat-style interface where users can ask basic questions about events, categories, or upcoming activities.

The project helps learners practice:

- Dynamic rendering using JavaScript
- State management on the client side
- Form handling and validation
- Filtering and searching
- CRUD operations
- Responsive layouts
- Chat-style UI interactions
- Local storage or mock data handling

Background

Many modern platforms such as meetup websites, workshop portals, and college fest systems allow users to discover and register for events easily. Users expect filtering, attendee visibility, and smooth interactions across devices.

This project simulates a real-world event platform where learners build multiple connected UI modules working together. The chat interface introduces learners to conversational UI patterns commonly used in modern applications.

Assumptions

- Event data can be stored using mock JSON or browser local storage.
- Authentication is not required.
- Registration does not require payment integration.
- Chat responses can be rule-based or simulated using predefined responses.
- Admin access can be simulated using a separate route/page/button.

Success Metrics / KPIs

Metric	Goal
Event registration flow completion	Users can register successfully without page refresh
Event filtering accuracy	Filters update listings correctly
CRUD operations	Admin can create, edit, and delete events successfully
Responsive experience	Layout works properly on mobile, tablet, and desktop
Chat usability	Users can send and receive chat messages smoothly

User Personas

Event Explorer

A user looking for interesting events based on interests and dates.

Event Organizer

A platform admin managing events and attendee visibility.

New User

A first-time visitor trying to discover events quickly using search, filters, and chat assistance.

Features Overview

Feature	Description	Priority
Event Listings	Display all available events in card/grid layout	MUST HAVE
Event Registration	Allow users to register for events	MUST HAVE
Attendee Count	Show number of attendees for each event	MUST HAVE
Date Filter	Filter events by selected date	MUST HAVE
Category Filter	Filter events by category	MUST HAVE
Event Details View	Expand or open event details	MUST HAVE

Feature	Description	Priority
Admin CRUD Panel	Create, update, and delete events	MUST HAVE
Homepage Chat Interface	Interactive chat widget on homepage	MUST HAVE
Responsive Design	Mobile-friendly layouts	MUST HAVE
Search Bar	Search events by title	GOOD TO HAVE
Dark Mode Toggle	Theme switching support	CAN BE DONE

User Stories

Event Browsing

As a user, I want to view all available events so that I can decide what interests me.

Event Registration

As a user, I want to register for an event so that I can attend it later.

Event Filtering

As a user, I want to filter events by date and category so that I can quickly find relevant events.

Attendee Visibility

As a user, I want to see attendee counts so that I can understand how popular an event is.

Admin Event Management

As an admin, I want to create, edit, and delete events so that event information stays updated.

Homepage Chat

As a visitor, I want to interact with a homepage chat assistant so that I can discover events more easily.

First-Time User

As a new user, I want a simple and guided homepage experience so that I can understand the platform quickly.

Functional Requirements

ID	Requirement	Acceptance Criteria	Priority
FR-1	Display event cards	Cards show title, image, category, date, location, attendee count	MUST HAVE
FR-2	Register button	Clicking register updates attendee count	MUST HAVE
FR-3	Date filtering	Events update dynamically based on selected date	MUST HAVE
FR-4	Category filtering	Users can filter events using dropdown/buttons	MUST HAVE
FR-5	Search functionality	Users can search events by event name	GOOD TO HAVE
FR-6	Admin create event	New event appears instantly in listing	MUST HAVE
FR-7	Admin edit event	Existing event data updates correctly	MUST HAVE
FR-8	Admin delete event	Event is removed from UI immediately	MUST HAVE
FR-9	Homepage chat UI	Users can send and receive messages	MUST HAVE
FR-10	Responsive layouts	UI adapts across device sizes	MUST HAVE

Homepage Chat Requirements

The homepage should include a floating or embedded chat interface.

The chat system should:

- Allow users to type messages
- Display messages in chat bubble format
- Support simple predefined responses
- Suggest events based on keywords
- Scroll automatically as messages increase
- Include timestamps for messages

Example interactions:

- “Show me tech events”
- “Any workshops this weekend?”
- “Music events available?”

The responses can be mocked using static logic or keyword matching.

Admin Dashboard Requirements

The admin panel should include:

- Event creation form
- Edit event functionality
- Delete event action
- Event list/table view
- Form validation

Suggested fields:

- Event title
- Description
- Category
- Date
- Time
- Venue
- Banner image URL
- Maximum attendees

Suggested Pages

Page	Purpose
Homepage	Event discovery and chat interface
Event Details Page	Detailed event information
Registration View	User registration confirmation
Admin Dashboard	Event CRUD operations

Suggested Components

Component	Purpose
Navbar	Navigation links
Event Card	Event preview display
Filter Panel	Category/date filters
Chat Widget	Homepage chat interface
Registration Button	Event registration
Admin Form	Create/edit events
Modal/Dialog	Confirm actions
Toast Notification	Success/error feedback

Data Structure Example

```
JavaScript
{
  id: 1,
  title: "Frontend Bootcamp",
  category: "Technology",
  date: "2026-06-20",
  location: "Community Hall",
  attendees: 120,
  description: "Hands-on frontend workshop"
}
```

Validation Requirements

- Event title cannot be empty
- Date must be valid
- Attendee count cannot go below zero
- Required fields should show validation messages
- Chat input should not allow empty submissions

Non-Functional Requirements

Requirement	Details
Performance	Event filtering should feel instant
Accessibility	Buttons and forms should be keyboard accessible
Responsiveness	Layout must work across screen sizes
Code Quality	Components and logic should remain modular
Maintainability	Event data and UI logic should remain organized

Error Handling

Scenario	Expected Behavior
Empty filter result	Show “No events found” message
Invalid form input	Show validation feedback
Empty chat submission	Prevent sending
Missing event image	Display placeholder image

Future Scope

Possible future improvements:

- Real authentication system
- Payment gateway integration
- Live chat powered by AI
- QR-based event tickets
- Calendar integration
- Notifications and reminders
- Real backend APIs

Non-Goals

The following are out of scope:

- Real-time multiplayer chat
- Backend database implementation
- Payment systems
- Email or SMS integrations
- Role-based authentication systems

Submission Guidelines

Learners should submit:

- Complete frontend source code
- Responsive implementation
- Properly structured folders
- README with setup instructions
- Screenshots or short demo recording